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Effect of membrane morphology on the initial rate of protein fouling during microfiltration

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Abstract

Protein fouling remains a major problem in the use of microfiltration for many bioprocessing applications. Experiments were performed to evaluate the effect of membrane morphology and pore structure on protein fouling using different track-etched, isotropic, and asymmetric microfiltration membranes. Fouling of membranes with straight-through pores occurred by pore blockage caused by deposition of large protein aggregates on the membrane surface. However, the rate of blockage was a function of the membrane porosity due to the possibility of multiple pore blockage by a single protein aggregate on high porosity membranes. Membranes with interconnected pores fouled more slowly since the fluid could flow around the blocked pores through the interconnected pore structure. This behavior was quantified using model membrane systems with well-defined pore morphology constructed from track-etch and isotropic membranes in a layered series combination. These results provide important insights into the effects of membrane pore structure and morphology on protein fouling. © 1999 Elsevier Science B.V. All rights reserved.

Keywords: Fouling; Microfiltration; Microporous and porous membranes; Membrane porosity; Membrane morphology

1. Introduction

Microfiltration is widely used for the purification of protein-containing solutions, both for initial clarification and sterile filtration. One of the critical issues in the development of effective microfiltration processes is the largely irreversible decline in system performance due to protein fouling arising from specific interactions between the proteins and membrane [1]. Several recent studies have demonstrated that the initial fouling during protein microfiltration is typically caused by the physical deposition of aggregated and/or denatured protein on the upper surface of the membrane. These studies have also provided important insights into the effects of solution pH [2–4], ionic strength [5,6], and specific chemical additives [2] on the extent and rate of protein fouling during microfiltration.

There is also extensive experimental evidence that protein fouling is strongly influenced by the physical and chemical characteristics of the microfiltration membranes, and much of this work has been discussed in the review paper by Marshall et al. [7]. Several studies have demonstrated that protein adsorption (in the absence of filtration) is reduced on very hydrophilic microfiltration membranes [8], although the filtrate flux during actual protein filtration is often

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found to be largely independent of the membrane surface chemistry [9,10]. For example, Mueller and Davis [10] found very similar flux decline profiles during constant pressure BSA filtration through 0.5 μm pore size polyethylene membranes both untreated and after treatment with polyvinylalcohol to render the membrane surface more hydrophilic. This behavior is consistent with previous studies showing that fouling occurs primarily by the physical deposition of large protein aggregates on the membrane surface. The reduction in protein adsorption on hydrophilic membranes has relatively little effect on the filtrate flux since monolayer adsorption causes little change in pore diameter due to the relatively large initial pore size of the microfiltration membranes [1].

1.1. Theoretical background

Most studies of protein fouling have interpreted the observed differences in flux decline during protein filtration using the classical pore blockage, pore constriction, and/or cake filtration models [11,12]. In the complete pore blockage model, the normalized flux is given as [12]:

$$\frac{J}{J_0} = \exp\left(-\frac{\alpha_1 J_0 A C_b}{N_0} t\right), \tag{1}$$

where J_0 is the initial filtrate flux through the clean membrane, A is the membrane surface area, C_b the bulk concentration, and N_0 is the initial pore density of the membrane. The parameter α_1 is equal to the number of pores blocked per unit mass of protein convected to the membrane and should thus be directly proportional to the fractional concentration of the fouling component(s), i.e. the protein aggregates, present in solution. The intermediate pore blockage model allows for superposition of particles on the membrane with the normalized flux given as [11]:

$$\frac{J}{J_0} = \left(1 + \frac{\alpha_1 J_0 A C_b}{N_0} t\right)^{-1}. \tag{2}$$

The corresponding expression for the normalized flux given by the pore constriction model is [12]:

$$\frac{J}{J_0} = \left(1 + \frac{\alpha_2 J_0 A C_b}{N_0 \pi r_0^2 \delta_m} t\right)^{-2}, \tag{3}$$

where r_0 is the initial pore radius and δ_m is the membrane thickness. The parameter α_2 equals the volume of foulant deposited in the pore interior per unit mass of protein filtered through the membrane. In the cake formation model [12]:

$$\frac{J}{J_0} = \left(1 + \frac{2\alpha_3 J_0 C_b}{R_m} t\right)^{-1/2}, \tag{4}$$

where R_m is the resistance of the clean membrane and α_3 is related to the rate of formation and the hydraulic resistance of the particle cake. A more detailed discussion of the underlying assumptions and mathematical development of these models is provided by Hermia [11] and Zeman and Zydney [12].

A very convenient approach for distinguishing between the predictions of these different fouling models is to evaluate the fouling parameter ω_0:

$$\omega_0 = \frac{K_0}{J_0 A C_b}, \tag{5}$$

where K_0 is the initial rate of flux decline

$$K_0 = -\frac{1}{J_0}\frac{dJ}{dt}\bigg|_{t=0}. \tag{6}$$

According to the pore blockage model

$$\omega_0 = \frac{\alpha_1}{N_0} = \frac{A_{pore}}{\epsilon A}\alpha_1, \tag{7}$$

with the same result obtained with the intermediate pore blockage model since at $t=0$ the probability of ''blocking'' an already blocked pore is negligible. The corresponding expressions for the pore constriction and cake formation models are:

$$\omega_0 = \frac{2\alpha_2}{N_0 \pi r_0^2 \delta_m}, \tag{8}$$

$$\omega_0 = \frac{\alpha_3}{A R_m}. \tag{9}$$

The fouling parameter is thus predicted to vary as $1/N_0$ (i.e. inversely proportional to the membrane porosity) for the pore blockage and pore constriction models but is independent of membrane porosity for the cake formation model. In contrast, ω_0 is predicted to vary inversely with the membrane thickness for the pore constriction model but is independent of δ_m for the other models.

1.2. Previous work

Bowen and Gan [13] obtained data for the flux decline during constant pressure filtration of bovine serum albumin (BSA) solutions through 0.2 μm pore size polycarbonate, anodized aluminum oxide (Anopore), and polyvinylidene fluoride (PVDF) membranes. The flux decline was much more pronounced for the polycarbonate membrane, which the authors attributed to the lower porosity (9% compared to 50% for the Anopore and 75% for the PVDF). Results were analyzed using the pore constriction model (Eq. (3)), with the data in good agreement with the linear relationship between total filtrate volume and reciprocal time predicted by this model. However, the best fit value of the parameter α_2 was unexpectedly much larger for the polycarbonate membrane. This behavior could not be directly explained in terms of the known difference in membrane properties, although Bowen and Gan hypothesized that it might be related to the different hydrodynamic conditions at the entry to, and inside of, the pores in the polycarbonate membrane due to its low porosity and smaller ratio of pore length to diameter.

Mueller and Davis [10] examined the fouling characteristics of 0.2 μm pore size polycarbonate, polysulfone, cellulose acetate, and PVDF membranes during constant pressure microfiltration of BSA. The extent of fouling was smallest for the cellulose acetate membrane, which they attributed to the much higher porosity of this microfiltration membrane. However, the flux decline for the polysulfone membrane was more rapid than that for the polycarbonate membrane, even though the polysulfone membrane had a slightly greater surface porosity (12% vs. 9%) and a much greater bulk porosity (65–80% compared to only 9% for the track-etch polycarbonate membrane). Mueller and Davis hypothesized that this difference might be related to the greater thickness and lower hydrophilicity of the polysulfone membrane, but no quantitative analysis of these effects was provided. A subsequent study by Güell and Davis [14] showed that lysozyme fouling was more rapid for the polycarbonate membrane than for the polysulfone, an effect which they attributed to the greater size of the lysozyme aggregates (compared to those formed by BSA). Data obtained with ternary mixtures of lysozyme, ovalbumin, and BSA showed a greater rate of fouling with lower porosity membranes, although the initial rate of flux decline was actually smallest for the PVDF membrane which has a surface porosity somewhat smaller than that for the cellulose acetate. The filtrate flux data in both studies were analyzed by plotting the total resistance as a function of time using the approach developed by Tracey and Davis [15]. The results suggested that there was an initial period of ''internal fouling'' (consistent with either pore blockage or pore constriction) followed by some type of ''external fouling'' (cake formation). The relative importance of these two mechanisms depended on the membrane, with the lower-fouling cellulose acetate and PVDF membranes showing primarily internal fouling while the polysulfone and polycarbonate membranes showed both internal and external fouling. These differences were attributed to the more open and mesh-like morphology of the CA and PVDF membranes [14].

Hlavacek and Bouchet [16] used constant flux experiments to examine the effects of membrane properties on protein fouling. The most rapid fouling (i.e. increase in pressure) was obtained with a polycarbonate membrane. The rate of pressure increase was actually smallest for the hydrophobic PVDF membrane, with slightly higher rates of fouling seen with both a hydrophilic cellulose membrane and a hydrophilic version of the PVDF. The pressure data were in good agreement with the intermediate pore blocking model, with the blocking coefficient α_1 a strong function of solution pH.

Although these studies have provided some general insights into the effects of membrane pore morphology on membrane fouling, it is still unclear exactly how the membrane porosity and pore structure actually influence the rate of fouling. The specific objectives of this study were to: (1) obtain quantitative data for the flux decline during constant pressure filtration of BSA through membranes having different, well-defined, pore structures, (2) characterize the membrane pore structure and extent of fouling using scanning electron microscopy, and (3) develop appropriate mathematical descriptions for the initial rate of flux decline that specifically account for the effects of membrane morphology on the fouling behavior.

2. Materials and methods

2.1. Protein solutions

Phosphate buffered saline solutions (PBS) consisting of 0.03 M KH_2PO_4, 0.03 M $Na_2HPO_4 \cdot 7H_2O$, and 0.03 M NaOH were prepared by dissolving preweighed quantities of the appropriate salts (Sigma, St. Louis, MO) in the desired volume of deionized water obtained from a Barnstead water purification system (Barnstead/Thermodyne, Dubuque, IA) with resistivity greater than 18 MΩ cm.

All experiments were conducted with solutions of BSA prepared by dissolving crystalline BSA powder (fraction V heat shock precipitated BSA, catalog number A7906, Sigma) in PBS which had first been prefiltered through a 0.2 μm pore size Gelman Supor-200 membrane (Gelman Science, Ann Arbor, MI) to remove large particulates. Solution pH was measured to within 0.01 pH units using an Acumet 915 pH meter (Fisher Scientific, Pittsburgh, PA) and adjusted to 7.4 by addition of NaOH if needed. Protein solutions were freshly prepared before each experiment and used within 8 h of preparation.

Dynamic light scattering (DLS) measurements on the BSA solutions were performed using a Nicomp 380 system (Particle Sizing Systems, Santa Barbara, CA) equipped with a 15 mW laser diode (λ=635 nm) and a high gain avalanche photodiode detector. The detector was aligned to measure the light scattered at a 20° angle to the incident beam to increase the sensitivity to the forward scattering (Fraunhofer diffraction) associated with the larger particles. DLS measures the solute/particle diffusion coefficient, with the particle diameter calculated using the Stokes–Einstein equation [17,18]. The particle size distribution was evaluated from data for the autocorrelation function collected over a 10 min time period using Nicomp multimodal software.

2.2. Membranes

Data were obtained with a variety of membranes as listed in Table 1. All the membranes are reported by the manufacturers to be hydrophilic (the polycarbonate, PES, and PVDF membranes are all chemically modified using proprietary surface treatment to render them hydrophilic while the cellulosic, aluminum oxide, and polyester membranes are naturally hydrophilic). The polycarbonate (PCTE), polyester, and aluminum oxide (Anopore) membranes all have straight-through non-interconnecting pores as shown in the scanning electron micrographs in Fig. 1. The PCTE and polyester membranes are made by a track-etch process yielding very uniform cylindrical pores [20]. The PCTE-L is a low density (low porosity) version produced by irradiating the base polycarbonate for a shorter period of time. The pores in the Anopore membrane are formed from anodically oxidized aluminum and are somewhat more irregular in shape, although the pore size distribution is still quite narrow [21].

The polyvinylidene fluoride (PVDF), cellulose acetate (CA), and mixed cellulose ester (MF) membranes have a fibrous network structure resulting in an isotropic (interconnected) pore morphology as shown in Fig. 2. These membranes have a fairly broad pore size distribution, with the pore size on the upper

Table 1
List of membranes used in this study

Membrane	Membrane material	Manufacturer	Rating[a] (μm)	Catalog number	Thickness[a] (μm)	Porosity[b] (%)
PCTE-L	Polycarbonate	Osmonics, CA	0.2	11014	10	3
PCTE	Polycarbonate	Osmonics, CA	0.2	11013	10	10
PETE	Polyester	Osmonics, CA	0.2	31023	10	10
Anopore	Anodized aluminum	Whatman, UK	0.2	2250	60	60
PVDF	Polyvinylidene fluoride	Millipore, MA	0.22	GVWP02500	125	70
MF	Mixed cellulose esters	Millipore, MA	0.22	GSWP02500	150	70 (32)
CA	Cellulose acetate	Sartorious, NY	0.2	SM11127	125	70 (14)
PES	Polyethersulfone	Filtron, MA	0.16	OM994025	0.5 (skin)	80

[a]The pore size rating and the thickness were provided by the manufacturer.
[b]Values shown are the bulk porosity provided by the manufacturer. The number in parenthesis is the surface porosity reported by Zeman [19].

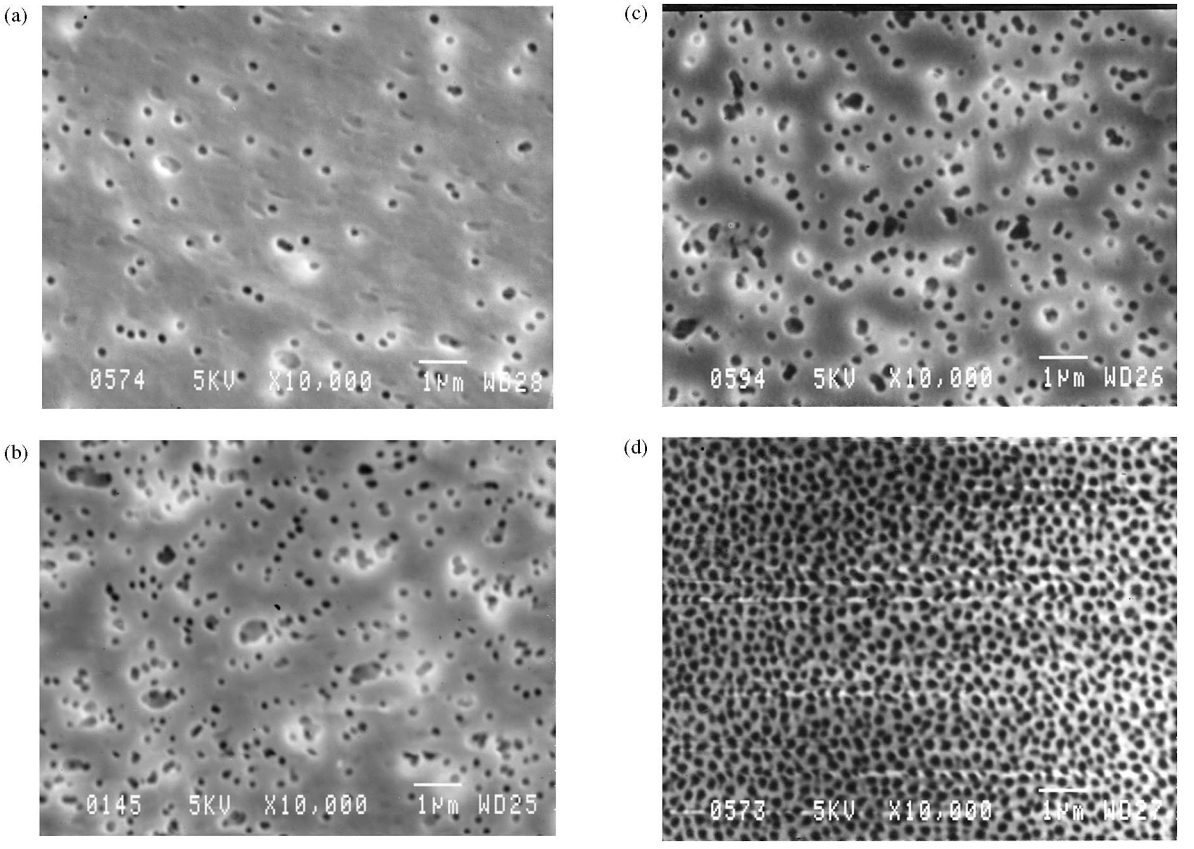


Fig. 1. Scanning electron micrographs of the surface of clean (a) low porosity PCTE, (b) high porosity PCTE, (c) polyester, and (d) Anopore membranes.

surface of the membrane being considerably larger than the rated (0.2 μm) pore size. The PES membranes are anisotropic consisting of: (a) an ultrathin skin which determines the sieving characteristics of the membrane, (b) a porous polyethersulfone substructure, and (c) a porous matrix made of Tyvek (polyethylene) which provides the membrane its structural integrity. The surface of the PES membrane has a much lower porosity than the bulk, with nearly circular interconnected pores in the range 0.1–1.5 μm.

To explore the effect of membrane morphology in more detail, experiments were also performed with the PCTE and PVDF membranes in ''sandwich'' combinations. Thus, the effect of membrane thickness was examined using either multiple PVDF membranes in series or ''composite'' membranes formed by placing a single PCTE membrane directly on top of one or more PVDF membranes. The membranes were obtained from the manufacturers as 25 mm disks and were placed directly in the base of the stirred cell. The PCTE and Anopore membranes were typically placed on top of a macroporous polyester drain disk (Osmonics, Livermore, CA) to minimize pore blockage by the support plate at the base of the stirred cell.

The surface characteristics of the clean and fouled membranes were also examined by scanning electron microscopy. Fouled membranes were prepared by first flushing with 100 ml of 0.2 μm-filtered PBS to remove any labile protein. The protein deposit was fixed by soaking the membrane in an 8% glutaraldehyde solution for 30 min. The membranes were then rinsed with PBS and dehydrated by repeated washing in ethanol solutions of successively increasing concentration, with the final washing done using 100% ethanol.

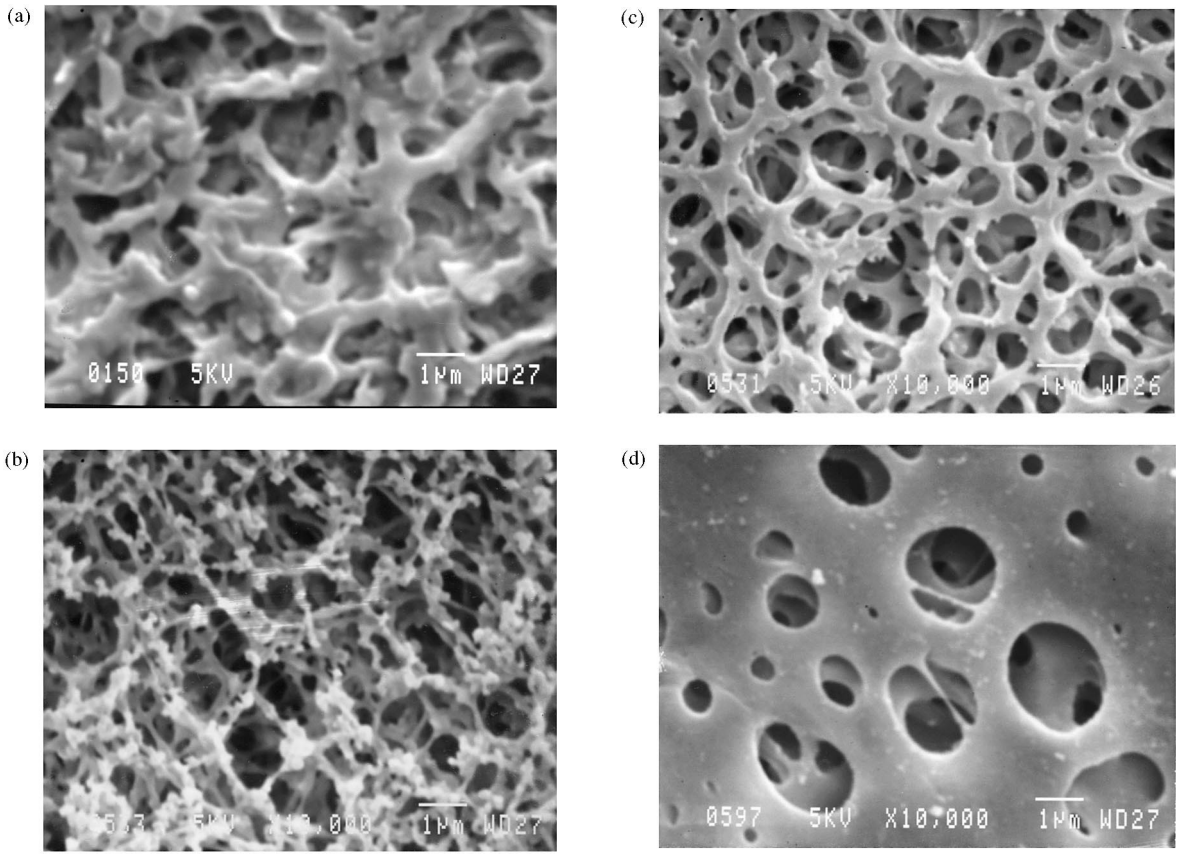


Fig. 2. Scanning electron micrographs of the surface of clean (a) polyvinylidene fluoride (PVDF), (b) mixed cellulose esters (MF), (c) cellulose acetate (CA), and (d) polyethersulfone (PES) membranes.

The specimens were air dried at 4°C, mounted on specimen studs, sputtered with a thin layer of gold under 25 kPa pressure using a Denton Vacuum Desk-1 etching/sputtering unit (Denton Vacuum, Moorestown, NJ), and imaged with a JEOL-840 scanning electron microscope (Jeol USA, Peabody, MA) at 5 kV.

2.3. Filtration experiments

All filtration experiments were performed in a 25 mm diameter stirred ultrafiltration cell (Model 8010, Amicon, Beverly, MA) connected to an acrylic solution reservoir. Constant pressure experiments were performed by air pressurizing the solution reservoir (from 14 to 120 kPa), with the filtrate flow rate measured by timed collection using a digital balance (Sartorious Model 1580, Edgewood, NY). The stirred cell and solution reservoir were initially filled with PBS. The membranes were then flushed with a minimum of 40 ml of PBS to remove any storage or wetting agents and insure adequate wetting of the pore structure. The PBS flux was measured at constant pressure until steady state was attained (usually within 30 min). The stirred cell was quickly emptied, refilled with BSA solution, and attached to a fresh reservoir containing additional BSA solution. The system was repressurized (within 1 min) and the stirrer speed set at 600 rpm (unless otherwise indicated) using a Strobotac Type 1531-AB strobe light (General Radio, Concord, MA). At the end of the filtration, the stirred cell was gently rinsed with PBS and the steady-state PBS flux re-evaluated. All experiments were conducted at room temperature (22±2°C).

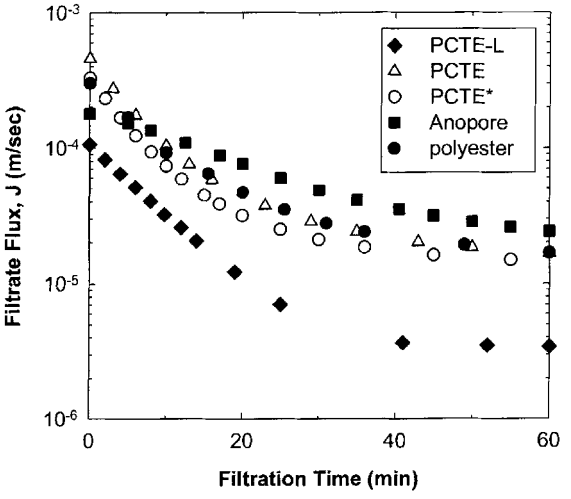


Fig. 3. Filtrate flux as a function of filtration time during filtration of 2 g/l BSA solutions through membranes with straight-through pores at a constant pressure of 14 kPa (2 psi). PCTE* is the polycarbonate membrane used without the drain disk.

3. Results

3.1. Straight-through pore membranes

Typical experimental data for the filtrate flux (J) during the constant pressure (ΔP=14 kPa=2 psi) filtration of 2 g/l BSA solutions through the straight-through pore polycarbonate (PCTE and PCTE-L), polyester, and aluminum oxide (Anopore) membranes are shown in Fig. 3. The initial flux through these membranes ranged from 1.2×10^{-4} to 4.6×10^{-4} m/s. In each case, this initial flux (measured within the first 30 s of the BSA filtration) was within 20% of the steady-state saline flux evaluated just prior to the start of the protein filtration. The lowest initial flux was attained with the PCTE-L membrane because of its very low porosity. The Anopore membrane also has a relatively low initial flux due to its greater thickness (60 vs. 10 μm for the PCTE and polyester membranes). The initial flux through the PCTE membrane without the drain disk was about 20% smaller than that with the drain disk. This difference was due to pore blockage at the base of the stirred cell caused by the support plate pressing directly against the underside of the membrane when used without the drain disk. Similar behavior was seen with the PCTE-L, polyester, and Anopore membranes. The filtrate flux during BSA filtration declined quite rapidly for all the membranes, with the flux decreasing by an order of magnitude within 20 min for the PCTE-L compared to more than 86 min for the Anopore membrane. The flux appears to attain a quasi-steady value at long filtration times, with the magnitude of the quasi-steady flux increasing with increasing membrane porosity.

In order to understand the basis for the different fouling behavior seen with the different straight-through pore membranes, the data in Fig. 3 were replotted in Fig. 4 as the normalized flux (J/J_0) as a function of the product J_0AC_bt which is consistent with the form predicted by the different classical fouling models (Eqs. (1)–(4)). Note that the linear dependence of $\ln(J/J_0)$ vs. J_0AC_bt seen at short times (r^2>0.99 for all runs using data for $J_0AC_bt<1.5\times10^{-4}$ kg) is exactly that predicted by the complete pore blockage model (Eq. (1)). The flux decline data for the equal porosity PCTE and polyester membranes were nearly identical when plotted in this manner, indicating that the specific surface chemistry of the membrane has little effect on the fouling in these experiments. This behavior is consistent with previous studies showing that the flux decline during BSA microfiltration is dominated by the physical deposition of large protein aggregates on the mem-

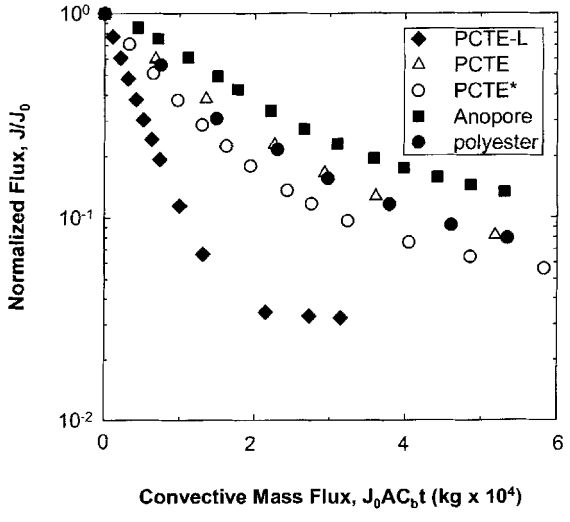


Fig. 4. Normalized filtrate flux as a function of J_0AC_bt during filtration of 2 g/l BSA solutions through membranes with straight-through pores at a constant pressure of 14 kPa (2 psi). PCTE* is the polycarbonate membrane used without the drain disk.

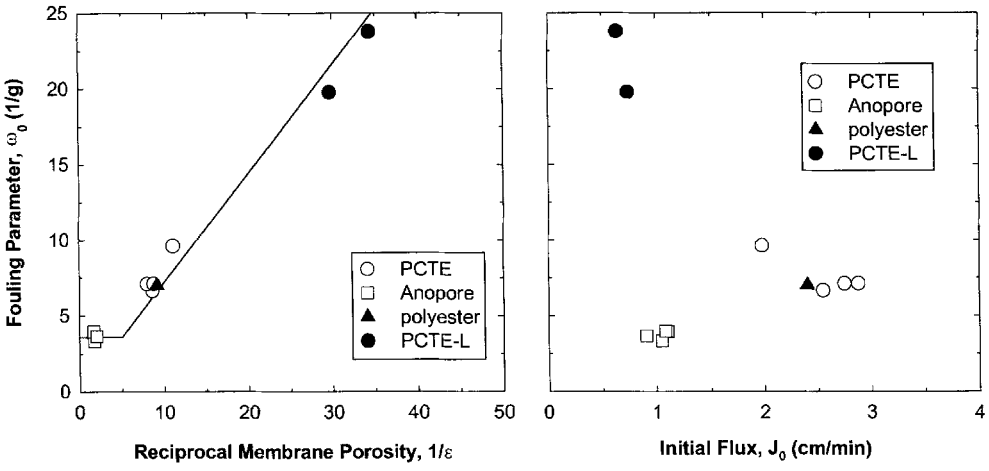


Fig. 5. Fouling parameter for membranes with straight-through pores plotted according to the pore blockage (left panel) and cake formation (right panel) models. The solid curve in the left panel is the model calculation as described in the text.

brane surface [22]. The rate of flux decline was greatest for the lowest porosity membrane, with the flux declining by 90% for the PCTE-L after $J_0C_bAt=10^{-4}$ kg compared to more than 6×10^{-4} kg for the high porosity Anopore membrane. However, the dependence on membrane porosity is nonlinear, with the polyester membrane ($\epsilon\approx0.1$) fouling only slightly more rapidly than the Anopore membrane ($\epsilon\approx0.6$) despite the factor of 6 difference in porosity.

The calculated values for the fouling parameter (ω_0) determined for the membranes with straight-through pores (both with and without the drain disk) are shown in Fig. 5. In each case, ω_0 was evaluated numerically directly from the initial slope of the $\ln(J/J_0)$ vs. J_0AC_bt data in Fig. 4:

$$\omega_0 = \frac{d\ln(J/J_0)}{d(J_0AC_bt)} \tag{10}$$

The left hand panel shows ω_0 as a function of $1/\epsilon$, which is the dependence predicted by the complete and intermediate pore blockage models, while the right hand panel shows ω_0 as a function of J_0, which is the dependence predicted by the cake formation model. The data are very well correlated with $1/\epsilon$ (consistent with the pore blockage models), while the results in the right hand panel are highly scattered. In particular, the fouling parameter for the Anopore membrane is a factor of 5 smaller than that for the low density PCTE, even though the initial flux through these membranes differ by less than 50%. The pore constriction model predicts that the fouling parameter should vary linearly with $1/(N_0\delta_m)$ (Eq. (8)) which is proportional to J_0/ϵ for runs at constant pressure. However, the PCTE, PCTE-L, and polyester membranes (both with and without the drain disk) all have the same J_0/ϵ since they have the same pore size and membrane thickness, while the actual data for ω_0 vary from 6.7 g^{-1} for the PCTE membrane to 24 g^{-1} for the low density PCTE.

Although the data in Fig. 5 are in good qualitative agreement with the pore blockage models, the data for the highest porosity membranes actually fall well above the predicted linear relationship between ω_0 and $1/\epsilon$. For example, the product $\omega_0\epsilon$ is essentially constant at $\omega_0\epsilon=0.67\pm0.02$ g^{-1} for the membranes with $\epsilon\leq0.1$, but $\omega_0\epsilon$ for the Anopore membrane is more than 1.8 g^{-1}. This suggests that pore blockage is in some way more effective, or at least more rapid, for the Anopore membrane than would be expected based on the results for the low porosity membranes. Our hypothesis is that this "enhanced fouling" is due to the possibility for multiple pore blockage by a single large aggregate on a high porosity membrane, an effect which is not accounted for in the simple pore blockage model.

For a low porosity membrane, where a given protein aggregate is able to block only a single pore, the

number of pores blocked per unit mass of material convected to the membrane surface (α_1) will be given as

$$\alpha_1 = \frac{f}{M_{agg}}, \tag{11}$$

where f is the fractional amount of BSA present in solution as aggregates and M_{agg} is the mass of a single aggregate. In contrast, for a high porosity membrane, each aggregate can block more than one pore, with the average number of pores/aggregate determined by the relative size of the aggregate and the pores as well as the pore spacing (i.e. the membrane porosity). These effects can be approximated as

$$\alpha_1 = \frac{f}{M_{agg}}\left(\frac{\epsilon A_{agg}}{A_{pore}}\right), \tag{12}$$

where A_{agg} is the surface area blocked by a single aggregate and A_{pore} is the cross-sectional area of a single pore. Thus, for a low porosity membrane ω_0 (as given by Eq. (7)) should proportional to $1/\epsilon$, but for a high porosity membrane ω_0 should become independent of the membrane porosity.

The solid line shown in the left hand panel of Fig. 5 is the model calculation given by Eqs. (7), (11) and (12) with the best fit value of $f/M_{agg}=9.1\times10^{9}$ g^{-1} determined by linear regression of the data for the PCTE, PCTE-L, and polyester membranes ($r^2=0.97$), and the best fit value of $A_{agg}/A_{pore}=5.6$ determined from the data points for the Anopore membrane (both with and without the drain disk). An independent estimate of A_{agg}/A_{pore} can be obtained from SEM micrographs of the fouled PCTE membranes (Fig. 6). The results suggest that $A_{agg}/A_{pore}=2$–8, in good agreement with the best fit value determined from the ω_0 data. An independent estimate of the fractional amount of BSA present in these protein solutions was obtained by dynamic light scattering, with a typical plot of the intensity of the scattered light at 20° shown in Fig. 7. The first peak occurs at a mean size of 8 nm, consistent with the average size of a BSA monomer. There is also a peak around 360 nm which corresponds to the aggregate. The intensity of the scattered light is a strong function of the particle size, with the large aggregates causing much more scattering, leading to the large size of the aggregate peak. The weight fraction of the protein aggregate can be calculated from the intensity data using the Nicomp multimodal software assuming that the monomer and the aggregates have similar absorption characteristics as $f=0.0003$. The best fit value of f/M_{agg} would thus correspond to $M_{agg}=3\times10^{-14}$ g. If we treat the aggregates as an effective hard sphere with specific gravity of one then this value of M_{agg} would correspond to an average diameter of about 0.4 μm, in excellent agreement with the average size of the aggregates observed by dynamic light scattering (Fig. 7) and on the SEMs (Fig. 6). Similar size aggregates were seen by Kelly and Zydney [2] for SEMs prepared by critical point drying. Thus, the modification of the pore blockage model to explicitly account for the effects of multiple pore blockage provides an accurate description of the fouling data obtained with all the different straight-through pore membranes.

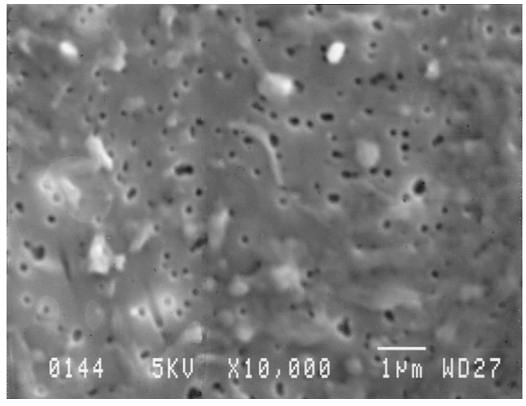


Fig. 6. Scanning electron micrographs of the surface of a PCTE membrane after filtering 2 g/l BSA solution for 10 min at a constant pressure of 2 psi.

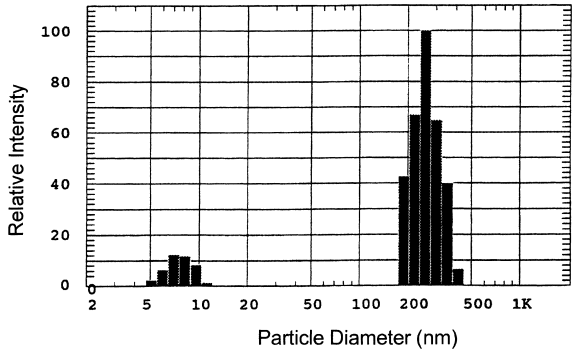


Fig. 7. Particle size distribution for the 2 g/l BSA solution used in this study.

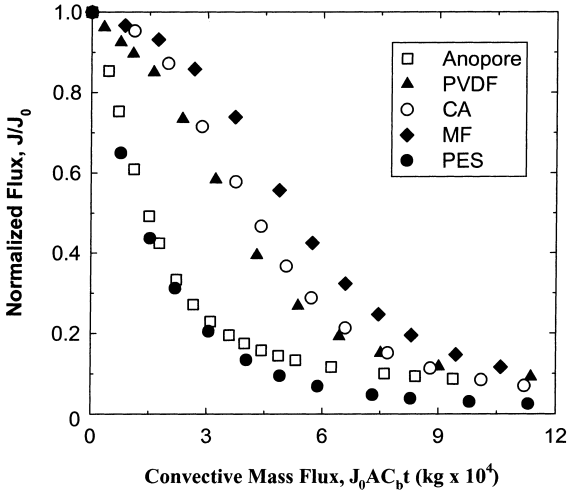


Fig. 8. Normalized filtrate flux for the filtration of 2 g/l BSA solutions through different 0.2 μm microfiltration membranes at a constant pressure of 14 kPa (2 psi).

3.2. Isotropic membranes

Fig. 8 compares the flux decline data for the constant pressure (ΔP=14 kPa=2 psi) filtration of 2 g/l solutions of BSA through the Anopore membrane with that for the isotropic polyvinylidene fluoride (PVDF), cellulose acetate (CA), and mixed cellulose esters (MF) membranes and also with data for the anisotropic polyethersulfone (PES) membrane. The surface pore structure of these latter membranes are shown in the scanning electron micrographs in Fig. 2. The filtrate flux data in Fig. 8 are again plotted as the normalized flux (J/J_0) as a function of J_0AC_bt, with the J_0 values summarized in Table 2. The MF membrane had a much larger J_0 value than the PVDF membrane even though these membranes have similar porosity, thickness, and nominal pore size. Note that the SEMs do show a much ''lacier'' structure for the mixed cellulose esters membrane, and this may have contributed to the larger initial flux. The flux data for the PES membrane were very similar to those for the Anopore membrane, and the reasons for this will be discussed in more detail subsequently. The rate of flux decline for the different isotropic membranes were considerably slower than that for the Anopore membrane. This difference is particularly striking for the Anopore and PVDF membranes which have similar porosity ($\epsilon \approx 0.7$) and initial filtrate flux ($J_0 \approx 2\times10^{-4}$ m/s) but very different rates of flux decline. This behavior is also seen in the magnitude of the fouling parameter, with ω_0=3.9 g^{-1} for the Anopore membrane compared to ω_0=0.4–0.9 g^{-1} for the isotropic membranes (values summarized in Table 2). Thus, these isotropic membranes, all of which have an interconnected pore structure, clearly have a much lower rate of flux decline than the membranes with straight-through pores, and this effect is not simply due to differences in membrane porosity and/or thickness.

Table 2
Initial flux, rate of flux decline, and fouling parameter evaluated for the different microfiltration membranes

Membrane	J_0 (m/s×10^4)	K_0 ($s^{-1}\times10^3$)	ω_0 (g^{-1})
PCTE-L	1.2	2.1	20.6
PCTE	4.6	2.7	7.1
Polyester	4.0	2.4	7.3
Anopore	1.8	0.6	3.9
PVDF	2.2	0.2	0.90
MF	5.8	0.2	0.41
CA	4.5	0.2	0.44
PES	4.4	2.0	5.5

These differences in fouling behavior can be understood by examining the fluid flow profile through an isotropic membrane as shown schematically in Fig. 9. In contrast to the behavior for a membrane with straight-through pores, the interconnected pore structure allows the fluid to flow around the protein aggregate as it passes through the membrane. Thus, the protein deposit only disturbs the fluid flow over a small penetration distance into the membrane. The net result is that pore ''blockage'' on an isotropic membrane would be expected to have a much smaller effect on the overall hydraulic resistance, and thus the filtrate flux, than that for a membrane with straight-through pores. This physical picture also explains why the flux decline data for the asymmetric polyethersulfone membrane in Fig. 8 looks so similar to that for the straight-through pore (Anopore) membrane. In this case, the very thin skin (0.5 μm thick) provides the

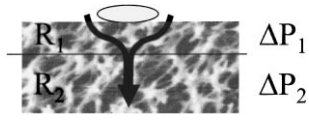


Fig. 9. Schematic diagram showing flow profile through a membrane with interconnected pores.

bulk of the resistance to flow in the asymmetric membrane, with any pore blockage on the low porosity upper surface disturbing the flow throughout the entire skin layer. Thus, the asymmetric membrane behaves as if it had non-interconnected pores that are fully "blocked" by any protein aggregate on the membrane surface.

In order to quantify this behavior, we have developed a simple (approximate) model for protein fouling that explicitly accounts for the connectivity of the pores. The hydraulic resistance of the isotropic membrane is approximated as the sum of the resistance of two distinct layers: a thin upper layer in which the flow is strongly disturbed by any pore blockage on the membrane surface and a thicker downstream layer in which the flow is essentially uniform irrespective of any disturbance (i.e. blockage) on the upper surface. The total volumetric flow rate through each layer must be equal:

$$Q = \frac{\Delta P_1}{\mu R_1}\frac{N_{\text{open}}}{N_0} = \frac{\Delta P_2}{\mu R_2}, \tag{13}$$

where ΔP_1 and ΔP_2 are the pressure drops across the two layers, and R_1 and R_2 are hydraulic resistances. Note that in writing the middle expression in Eq. (13) we have implicitly assumed that the upper layer can be approximated as having straight-through pores which are totally blocked by the aggregate. Eq. (13) can be solved for the volumetric flow rate in terms of the total transmembrane pressure drop ($\Delta P = \Delta P_1 + \Delta P_2$) yielding

$$Q = \frac{\Delta P}{\mu}\left(R_1\frac{N_0}{N_{\text{open}}} + R_2\right)^{-1}. \tag{14}$$

The fouling parameter for this type of isotropic membrane can be evaluated from Eq. (5) using the pore blockage model as

$$\omega_0 = -\left(\frac{R_1}{R_1+R_2}\right)\frac{1}{J_0 A C_b N_0}\frac{dN_{\text{open}}}{dt} = \frac{R_1}{R_1+R_2}\frac{\alpha_1 A_{\text{pore}}}{\epsilon A}. \tag{15}$$

Comparison of Eqs. (15) and (7) shows that the presence of an interconnected pore structure reduces the rate of fouling (i.e. the value of the fouling parameter) by the ratio of $R_1/(R_1+R_2)$ since the flow through the bulk (downstream layer) of the isotropic membrane is assumed to be unaffected by the pore blockage on the upper surface of the membrane. Comparison of the data for the Anopore and PVDF membranes yields $R_1/(R_1+R_2)=0.2$, while the values for the CA and MF membranes give $R_1/(R_1+R_2)$ closer to 0.1.

3.3. Sandwich configurations

In order to verify the dependence of the fouling parameter on the membrane resistance (i.e. on the ratio of $R_1/(R_1+R_2)$), constant pressure filtration experiments were performed using multiple PVDF membranes in series using a sandwich configuration. These experiments were performed without stirring to eliminate any artifacts associated with the change in distance between the upper surface of the membrane and the stirrer caused by an increase in thickness of the "composite" membranes. Previous studies [22] have shown that the initial rate of flux decline (K_0) during BSA microfiltration is independent of the stirrer speed under comparable experimental conditions. Fig. 10 shows data for the normalized flux as a function of filtration time for experiments performed with one, two, and four PVDF membranes stacked in series. In each case, the transmembrane pressure was adjusted to achieve an initial filtrate flux of $4.8\pm0.1\times10^{-4}$ m/s; thus, the pressure for the run with four PVDF membranes was almost exactly four times that for the run

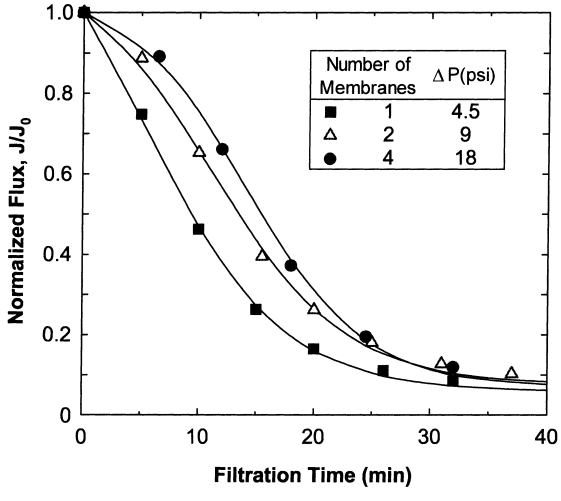


Fig. 10. Normalized filtrate flux during filtration of 2 g/l BSA solutions through series combinations of PVDF membranes with $J_0=4.8\pm0.1\times10^{-4}$ m/s.

with only a single membrane. The rate of flux decline was slowest for the run with four PVDF membranes; thus, increasing the membrane thickness actually reduced the rate of fouling. There was no evidence of any fouling within the subsequent PVDF membranes. Water flux measurements obtained with the individual membranes removed from the composite structure showed identical hydraulic permeability before and after the protein filtration. The absence of any fouling was also confirmed by scanning electron microscopy, which showed that all the aggregates were deposited on the upper surface of the uppermost membrane. Instead, the decrease in fouling was due to the increase in the overall resistance to flow (R_1+R_2) caused by the additional PVDF membranes. Note that R_1 should be unaffected by the addition of one or more membranes since the flow disturbance caused by aggregate deposition should only penetrate a few microns into the upper membrane (compared to a total thickness of 120 μm for each PVDF membrane).

Fig. 11 shows the results for the fouling parameter (ω_0) evaluated from the filtrate flux data in Fig. 10. The data are plotted as a function of the reciprocal of the membrane resistance $(R_1+R_2)^{-1}$ which was determined from saline flux measurements obtained for the composite membrane immediately prior to the BSA filtration. The data are clearly linear when plotted in this manner (r^2>0.94) with a slope of 7.7×10^{-8} $g^{-1}\,cm^{-1}$. Thus, the interconnected pore structure of the PVDF membranes has a significant effect on fouling, causing the rate of flux decline to decrease as the membrane thickness (or resistance) increases.

In order to obtain additional insights into the role of membrane morphology on protein fouling, a series of filtration experiments were performed using a model membrane system consisting of a single PCTE membrane on top of one or more PVDF membranes to form a single stacked membrane structure. Fig. 12 shows the normalized flux through these composite membranes as a function of filtration time, with the transmembrane pressure drop again adjusted to obtain the same initial filtrate flux (within 8%). The rate of flux decline is clearly lower for the composite having the most PVDF membranes. In addition, the time required to reach a quasi-steady flux increased with increasing number of underlying PVDF membranes. Again, scanning electron microscopy and hydraulic permeability measurements confirmed that all the fouling was localized on the upper surface of the PCTE membrane.

The data in Fig. 12, along with corresponding data obtained with low density PCTE membranes in series arrangement with the PVDF membranes, were used to evaluate the fouling parameter for these composite membrane structures. The results are shown in Fig. 13, with the data plotted as a function of R_1/R_{total} based on the expression for α_1 for a low porosity

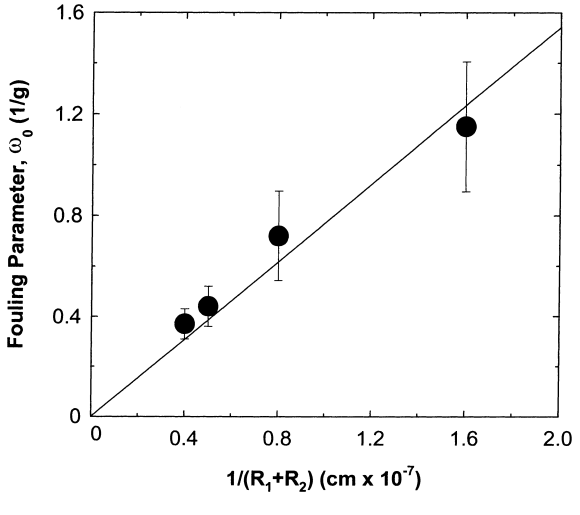


Fig. 11. Fouling parameter vs. the reciprocal membrane resistance for the stacked PVDF membranes.

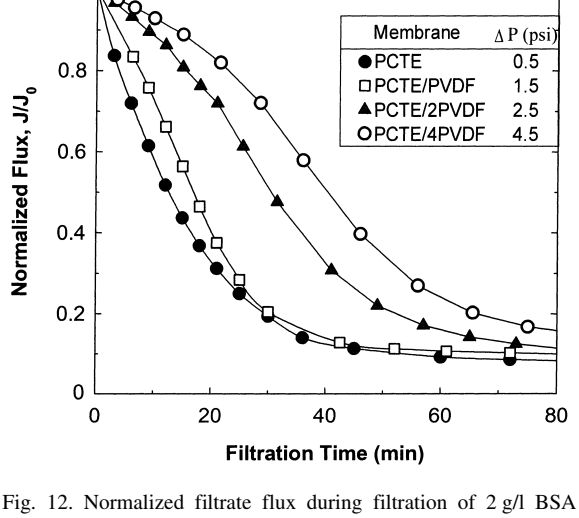


Fig. 12. Normalized filtrate flux during filtration of 2 g/l BSA solutions through PCTE/PVDF composite membranes with $J_0=1.7\pm0.1\times10^{-4}$ m/s.

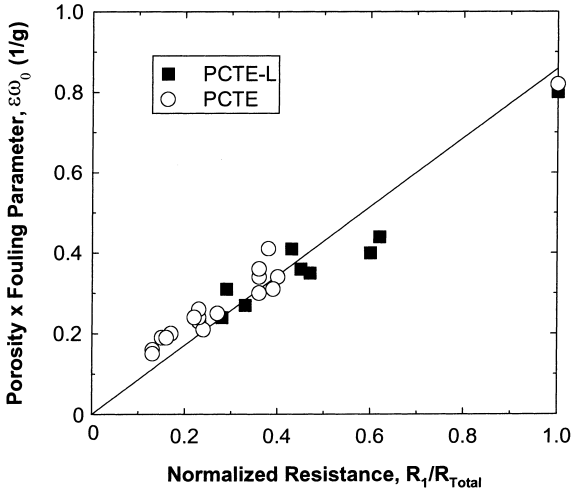


Fig. 13. Fouling parameter as a function of the ratio of upper layer to total resistance for experiments with composite PCTE/PVDF membranes.

membrane (Eq. (11)). In this case, R_1 was assumed to be equal to the resistance of the clean PCTE or PCTE-L membrane, with $R_{total}=R_1+R_2$ evaluated from the overall resistance of the composite structure as determined from water permeability measurements prior to the BSA filtration. The data are highly linear when plotted in this manner, providing further confirmation of the effects of porosity and pore connectivity on the flux decline.

The pore connectivity also has a large effect on the flux through more heavily fouled membranes. Table 3 shows data for the saline flux through clean and fouled PCTE and PVDF membranes. Fouling was done by constant pressure filtration of a 2 g/l BSA solution, with the initial transmembrane pressure adjusted to give the same initial flux (ΔP=1 psi for the PCTE and 2 psi for the PVDF membrane). The BSA filtration

Table 3
Saline flux for clean and fouled PCTE and PVDF membranes in normal and reversed orientations

Membrane orientation	Saline flux ($m/s\times10^4$)	
	PCTE	PVDF
Clean	2.8±0.3	2.8±0.3
Fouled	1.7±0.1	1.7±0.1
Reverse	1.7±0.1	2.5±0.1
Normal	1.7±0.1	1.7±0.1

was continued until J/J_0=0.6, at which point the membrane was gently rinsed with PBS to remove any labile protein. The saline flux through the fouled membranes were 40% smaller than the flux through the clean membranes, indicating that the flux decline seen during BSA filtration was due entirely to irreversible fouling and not to any concentration polarization (boundary layer) effects. The membranes were then removed from the stirred cell, flipped upside down, and returned to the stirred cell with the protein deposit now on the downstream surface of the membrane. The saline flux through the reversed PCTE membrane was essentially identical to that obtained in the original orientation, consistent with the blockage of a certain fraction of the membrane pores. In contrast, the saline flux for the reversed PVDF membrane was significantly larger than that measured in the original orientation. The two membranes were then flipped again, returning them to their original orientations. The saline flux through each membrane was very similar to that evaluated immediately after the BSA filtration; thus, the increase in flux seen for the reversed PVDF membrane was not simply due to partial removal of the protein deposit from the membrane surface. Instead, this increase in saline flux appears to reflect a true reduction in the effective resistance of the fouled PVDF membrane when it is operated with the protein deposit at the downstream surface of the membrane. This dependence of the resistance on membrane orientation is likely due to the difference in flow profiles when the fluid flows around the deposit first (i.e. as it enters the membrane in the original orientation) or last (i.e. in the reverse orientation). This effect was not seen with the PCTE membrane due to the straight-through pore structure of the track-etch membrane.

The effects of this internal flow distribution on membrane fouling were examined by performing constant pressure (ΔP=14 kPa=2 psi) filtration experiments in which the membrane orientation was reversed when the flux had dropped to about 70% of its initial value. Results are shown in Fig. 14 for both a PCTE and PVDF membrane. The flux decline at the start of the filtration was similar to that seen in Fig. 3 for the PCTE membrane and in Fig. 8 for the PVDF membrane, with J/J_0=0.7 obtained after 4 min of filtration for the PCTE membrane and 33 min for the PVDF. At this point, the membranes were gently

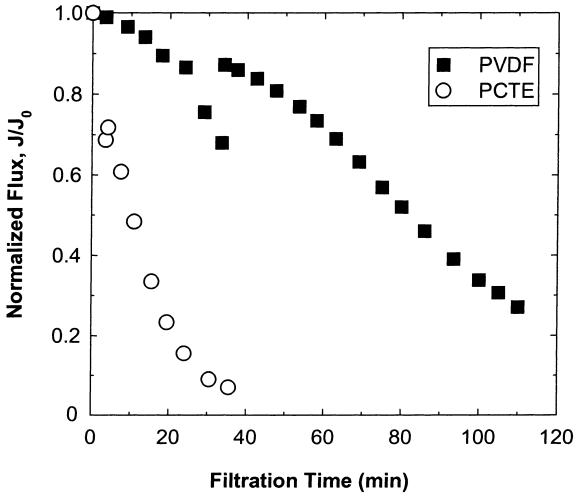


Fig. 14. Normalized flux during filtration of 2 g/l BSA solutions at 14 kPa (2 psi) through PCTE and PVDF membranes. Membrane orientation was reversed at $J/J_0 \approx 0.7$ for both membranes.

rinsed in PBS, removed from the stirred cell, flipped over, and returned to the stirred cell. The system was rapidly refilled with a fresh protein solution and re-pressurized to 2 psi. The flux decline profile through the PCTE membrane was largely unaffected by this change in membrane orientation, consistent with the view that the effects of pore blockage are independent of flow direction in a membrane with straight-through pores. The results for the PVDF membranes are markedly different. The normalized flux jumped from $J/J_0=0.68$ to $J/J_0=0.87$ upon reversing the membrane due to the reduction in effective membrane resistance in the reversed orientation. In addition, the rate of flux decline after the membrane reversal was much smaller than that measured prior to the reversal. In fact, the rate of flux decline through the reversed membrane was nearly identical to that measured at the start of the original filtration. In this case, the membrane reversal restored the upper surface of the membrane to a "clean" state, with aggregate deposition now occurring over the entire membrane surface and not just over the surface that remained aggregate-free.

4. Conclusions

The results obtained in this study provide the most complete description of the effects of membrane pore morphology, and in particular the membrane porosity and pore connectivity, on protein fouling. For membranes with straight-through pores, the initial rate of flux decline (as characterized by the fouling parameter ω_0) varied inversely with the membrane porosity at low porosities, but became independent of the porosity at higher porosities. An increase in porosity initially causes a reduction in the rate of flux decline since there is less convective flow through each pore of the membrane. However, as the membrane porosity gets even larger, it becomes possible for a single protein aggregate to effectively block multiple membrane pores. Under these conditions, the rate of flux decline becomes independent of the membrane porosity, in good agreement with data obtained for the high porosity Anopore membranes.

The rate of flux decline for membranes with interconnected pore structures was much smaller than that for membranes with straight-through pores, even for membranes with similar porosity. This reduction in the rate of flux decline occurred because the interconnected pore structure allowed the flow to go around the protein deposit as it passed through the membrane. Thus, the protein aggregate only disturbed the fluid flow over a small penetration distance into the membrane. Model calculations and experimental data for the sandwich membrane configurations clearly demonstrate that the membrane thickness can have a significant effect on the flux decline, even when all the fouling occurs on the uppermost surface of the membrane, due to the change in the fluid flow profiles throughout the overall membrane structure. A more complete analysis of the flow through membranes having this type of interconnected pore structure will be the subject of a future study.

The data obtained with the asymmetric polyethersulfone membrane provide a particularly interesting demonstration of the importance of the membrane pore structure/morphology on fouling. The flux decline for the asymmetric membrane was similar to that for the Anopore membrane, even though the SEMs clearly show an irregular and interconnected pore structure. However, in this case the upper selective layer of the membrane is so thin that the flow has insufficient time to pass around the pore blockage as it moves through the skin of the membrane, with the net result that the asymmetric membrane behaves as if it had non-connected straight-through pores.

Although the data obtained in this study focused primarily on the initial rate of flux decline, the pore morphology also has a strong effect on the longer term fouling behavior. This was seen very clearly in the data for the flipped PCTE and PVDF membranes, with the flow through the partially fouled PVDF membrane displaying a clear directional-dependence. The different flow distributions within the porous structure of these membranes can also alter the rate of subsequent protein deposition, thereby changing the approach to steady state. Finally, the quasi-steady flux appears to be a function of the membrane porosity (long time data in Fig. 3), but it is relatively independent of the pore connectivity, e.g. the steady flux through the Anopore and PVDF membranes were essentially identical. The underlying physical basis for these phenomena remains to be identified, but the results do suggest that a proper understanding of the effects of membrane porosity and pore morphology on protein fouling could be used in the development of new low-fouling membrane structures.

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